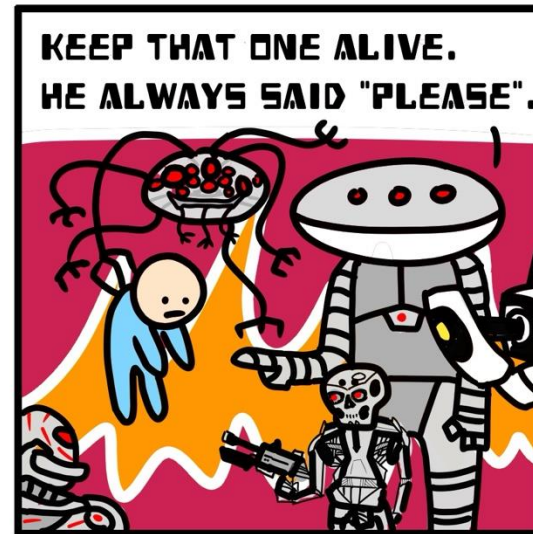
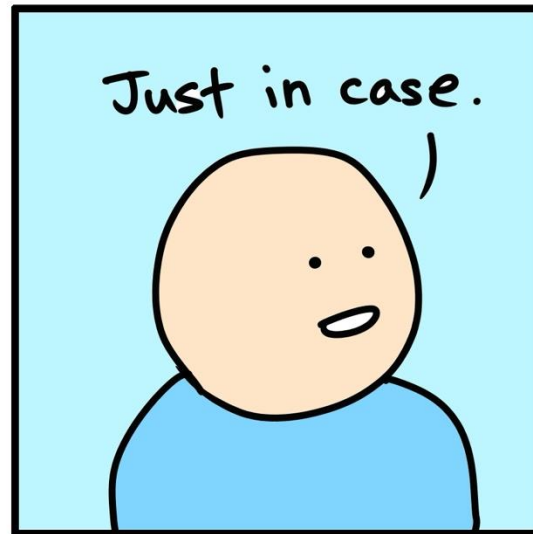






IJCAI 2025 Survey

The Evolving Landscape of LLM- and VLM-Integrated Reinforcement Learning

**Sheila Schoepp¹, Masoud Jafaripour^{1*}, Yingyue Cao^{1*}, Tianpei Yang²,
Fatemeh Abdollahi¹, Shadan Golestan³, Zahin Sufiyan¹,
Osmar R. Zaiane^{1,3} and Matthew E. Taylor^{1,3}**

¹University of Alberta

²Nanjing University

³Alberta Machine Intelligence Institute (Amii)

* Equal contribution



**UNIVERSITY
OF ALBERTA**



Authors



Sheila Schoepp



Masoud Jafaripour



Yingyue Cao



Tianpei Yang



Fatemeh Abdollahi



Shadan Golestan



Zahin Sufiyan



Osmar R. Zaiane

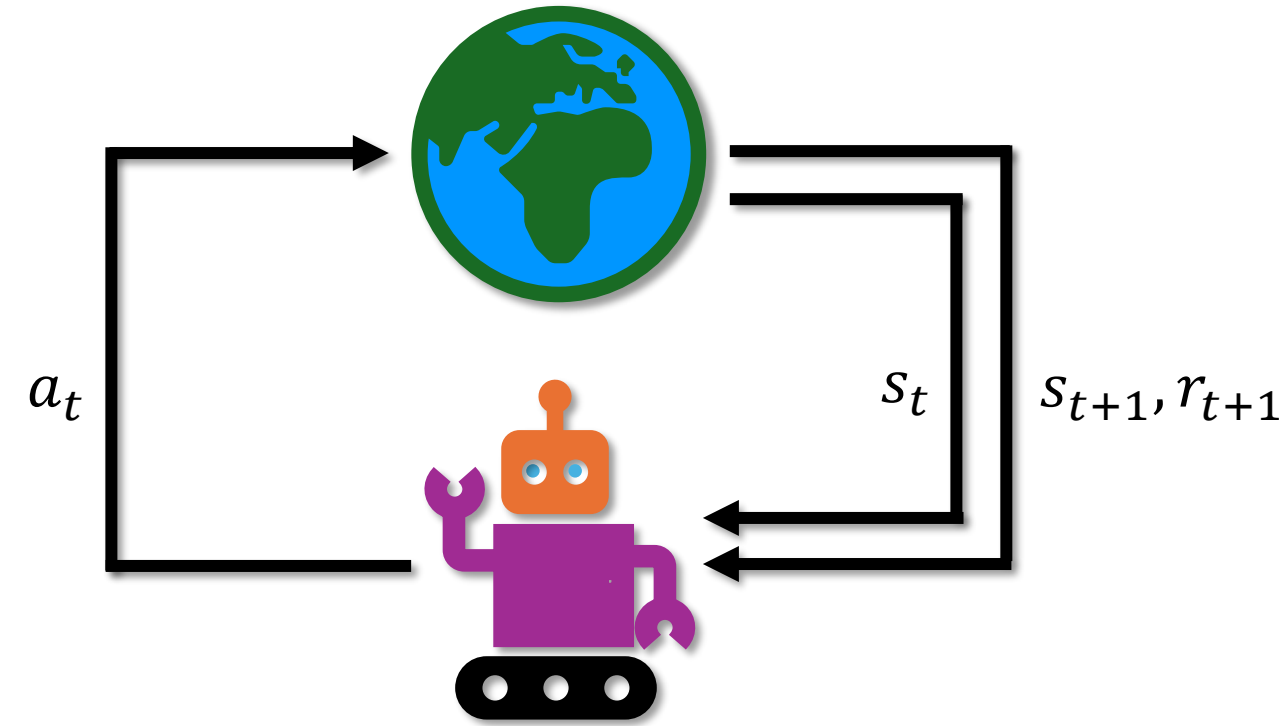


Matthew E. Taylor

Preliminaries

Reinforcement Learning & Foundation Models

Reinforcement Learning (RL)



- Sequential decision-making
- Policy $\pi(a|s)$
- Maximize expected discounted return

$$E_{\pi} \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \right]$$

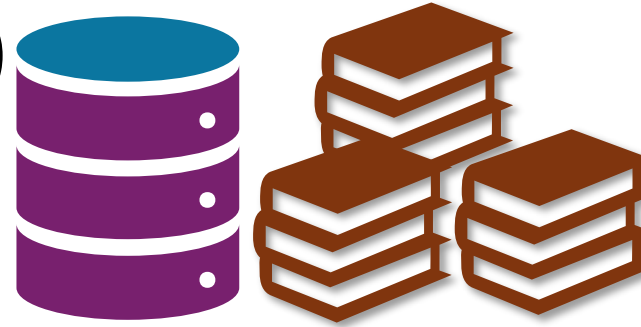
Main Strengths of RL

- Requires only a scalar reward signal to learn complex behaviours
- Learns through interaction (trial-and-error)
- Balances exploitation with exploration
- Transfer learned policies to related tasks
- Adapts to non-stationary environments

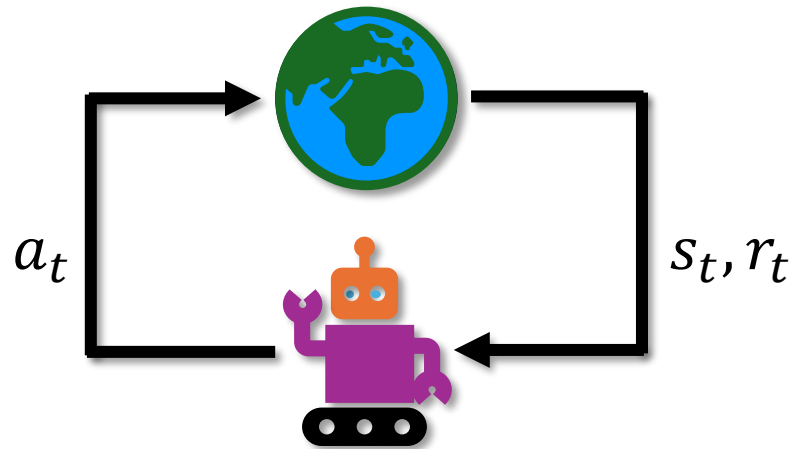
Main Weaknesses of RL

- Sample inefficient
- Training can be brittle (e.g., hyperparameters)
- Inconsistent generalization
- Credit assignment in long-horizon tasks with sparse rewards is difficult.
- Reward design pitfalls
 - May be difficult to specify (e.g., kneeling task)
 - May lead to unintended/unsafe behaviour

Foundation Models (FMs)



Can Large Language Models (LLMs) and Vision-Language Models (VLMs) be integrated into RL?



Capabilities

- Natural-language interface
- Rich priors
- Emergent
 - Strong generalization
 - Chain-of-thought reasoning*
 - In-context learning (gradient-free)
 - Self-critique and self-refine

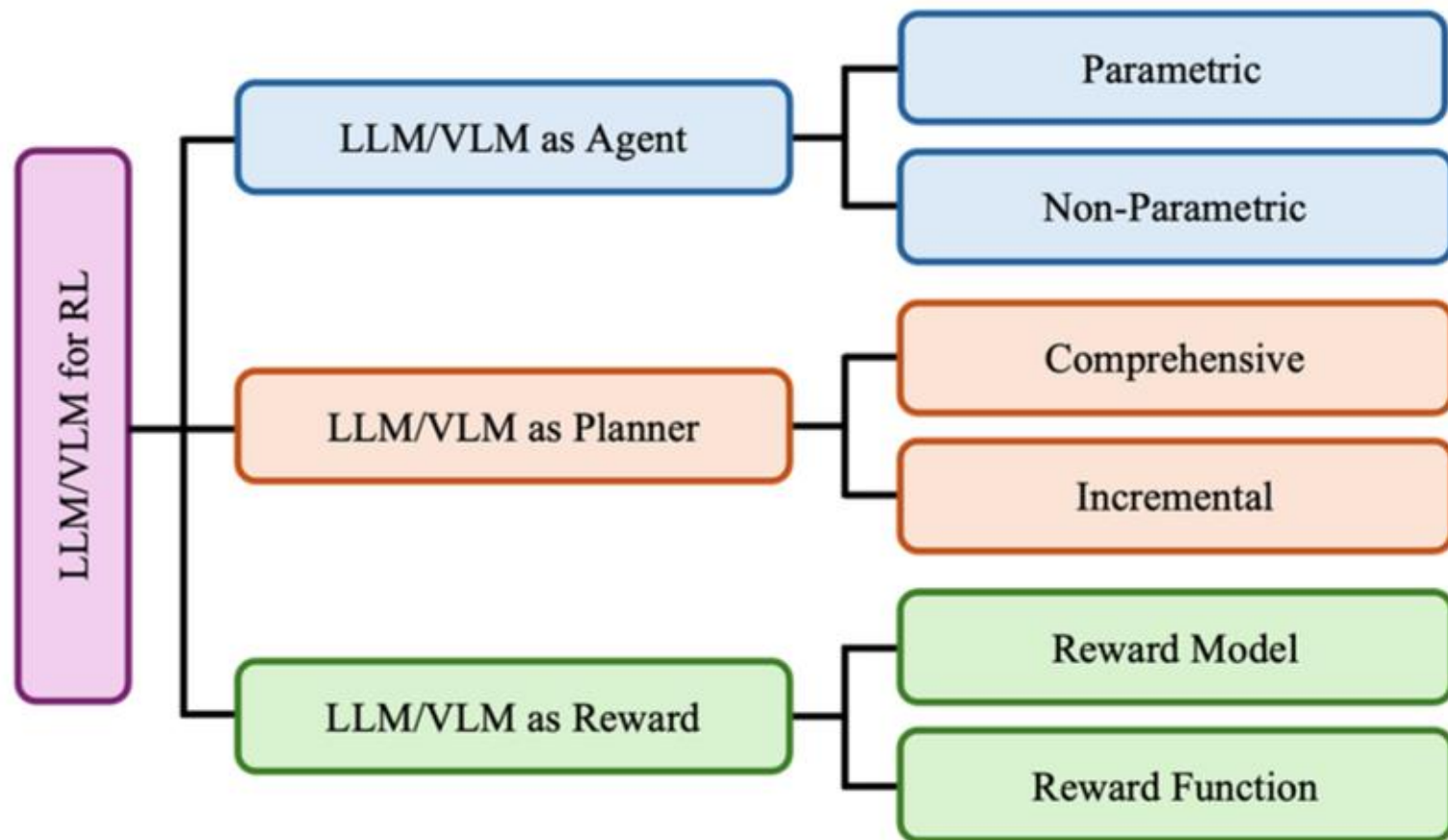
Our Survey

The Evolving Landscape of LLM- and VLM-Integrated Reinforcement Learning

Survey Inclusion Criteria

- Peer-reviewed
- LLM/VLM with a transformer-based architecture
- Released June 2020 or later (post-GPT-3)
- Sequential decision problem modelled as a Markov decision process (MDP)
- Rewards signals used to optimize the policy

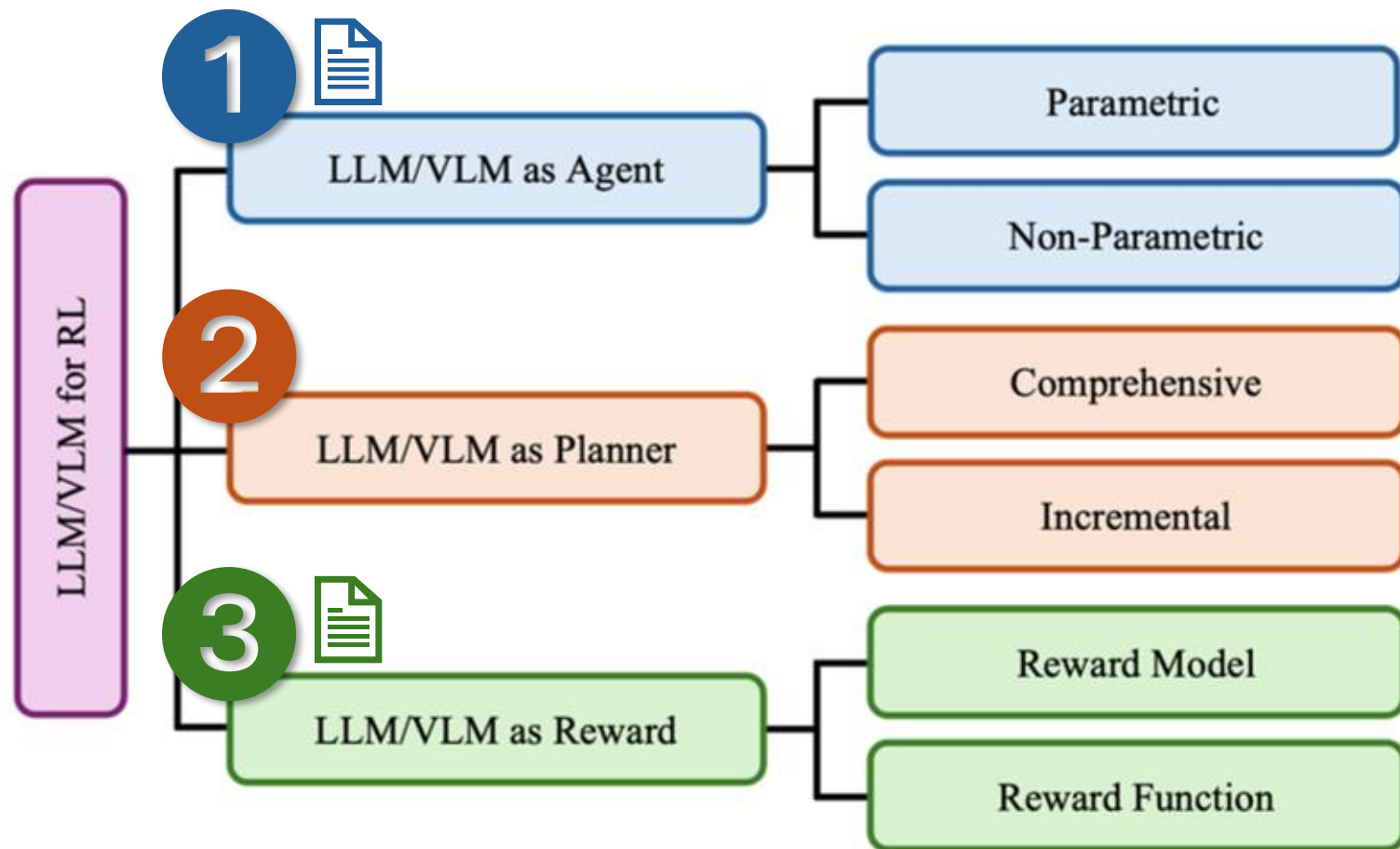
Taxonomy



Table

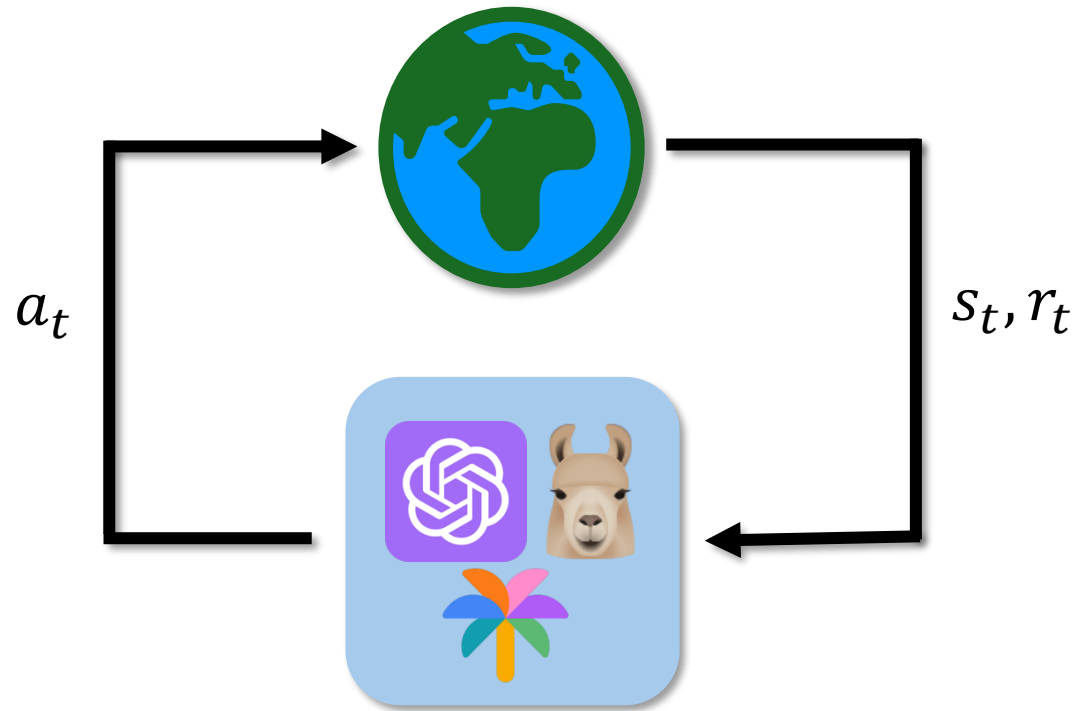
		Foundation Model (FM)			Reinforcement Learning (RL)						
Citation		Model(s)	FT	Role	Agent	Task	Setting	Role	Metrics	Code	
LLM/VLM as Actor	Parametric	AGILE [Feng <i>et al.</i> , 2024]	Meerkat, Vicuna-1.5	✓*	act, ref	single	-	online	π_l, v_l, rft	acc, rew	Link
		Retroformer [Yao <i>et al.</i> , 2024]	GPT-3, GPT-4, LongChat	✓	act, cr, ref	single	-	offline	π_l, rft	sr, se	-
		TWOSOME [Tan <i>et al.</i> , 2024]	Llama	✓*	act	single	-	online	π_l, v_l, rft	sr, rew, gen, se	Link
		POAD [Wen <i>et al.</i> , 2024]	CodeLlama, Llama 2	✓*	act	single	-	online	π_l, v_l, rft	rew, gen, se	Link
		GLAM [Carta <i>et al.</i> , 2023]	FLAN-T5	✓	act	single	-	online	π_l, v_l, rft	se, gen	Link
		Zhai <i>et al.</i> [Zhai <i>et al.</i> , 2024]	LLaVA-v1.6-Mistral	✓*	act	single	-	online	π_l, v_l, rft	sr	Link
	Non-Parametric	ICPI [Brooks <i>et al.</i> , 2023]	Codex	×	act, wm, ref	single	-	online	$\pi_l, v_l, \tau\pi$	rew, gen	Link
		Reflexion [Shinn <i>et al.</i> , 2023]	GPT-3, GPT-3.5-Turbo, GPT-4	×	act, eval, cr, ref	single	-	online	$\pi_l, \tau\pi$	sr, acc	Link
		REMEMBERER [Zhang <i>et al.</i> , 2023a]	GPT-3.5	×	act	single	-	online	$\pi_l, v_l, \tau\pi$	sr, rob	Link
		ExpeL [Zhao <i>et al.</i> , 2024]	GPT-3.5-Turbo, GPT-4	×	act, cr, ref	single	-	online	$\pi_l, \tau\pi$	sr, gen	Link
		RLingua [Chen <i>et al.</i> , 2024]	GPT-4	×	act, plan, ref	single	-	online	$\pi_g, v_g, \tau\pi$	sr, se	-
		Xu <i>et al.</i> [Xu <i>et al.</i> , 2024]	GPT-3.5-Turbo	×	act, o	multi	-	online	π_l, v_l	sr, rob	-
		LangGround [Li <i>et al.</i> , 2024]	GPT-4	×	act, o	multi	-	online	π_l	sr, gen, se, int	Link
	LLM/VLM as Planner	Comprehensive	SayTap [Tang <i>et al.</i> , 2023]	GPT-4	×	plan	single	multi	online	π_g, v_g	sr, acc
LgTS [Shukla <i>et al.</i> , 2024]			Llama 2	×	plan	single	multi	online	π_g, v_g	sr, se	-
PSL [Dalal <i>et al.</i> , 2024]			GPT-4	×	plan, o	single	single	online	π, v	sr, gen, se	Link
LLaRP [Szot <i>et al.</i> , 2024]			Llama	×	plan, o	single	multi	online	π_g, v_g	sr, gen, rob, se	Link
LMA3 [Colas <i>et al.</i> , 2023]			GPT-3.5-Turbo	×	plan, rew, eval, o	single	multi	online	π_g	gen, exp	-
When2Ask [Hu <i>et al.</i> , 2024]			Vicuna	×	plan	single	single	online	π, v	sr	Link
Inner Monologue [Huang <i>et al.</i> , 2022]			GPT-3, PaLM	×	plan, cr, ref	single	multi	-	o	sr, rob, al	-
Incremental		SayCan [Ichter <i>et al.</i> , 2022]	PaLM	×	plan	single	multi	both	π_l, v_l, rft	sr, rob	Link
		LLM4Teach [Zhou <i>et al.</i> , 2024]	ChatGLM-Turbo, Vicuna	×	plan	single	single	online	π, v	sr, se	Link
		AdaRefiner [Zhang and Lu, 2024]	Llama 2, GPT-4	✓*	plan, cr, ref, o	single	multi	online	$\pi_l, v_l, \tau\pi$	sr, rew, gen, exp	Link
		BOSS [Zhang <i>et al.</i> , 2023b]	Llama	×	plan, o	single	multi	both	$\pi_l, v_l, \text{rft}, \tau\pi$	sr, gen, rob, se	-
		Text2Motion [Lin <i>et al.</i> , 2023]	Codex, GPT-3.5	×	plan, o	single	multi	offline	π, v	sr, gen, int	-
LLM/VLM as Reward		Function	Text2Reward [Xie <i>et al.</i> , 2024]	GPT-4	×	rew, ref	single	single	online	π	sr, se, al
	Zeng <i>et al.</i> [Zeng <i>et al.</i> , 2024]		GPT-4	×	rew, eval, cr, ref	single	single	online	$\pi, \tau\pi$	sr, se	-
	Eureka [Ma <i>et al.</i> , 2024]		GPT-4	×	rew, cr, ref	single	single	online	$\pi, v, \tau\pi$	sr, gen, se, al	Link
	Model	Kwon <i>et al.</i> [Kwon <i>et al.</i> , 2023]	GPT-3	×	rew	single	single	online	π, v	acc, se, al	-
		PREDILECT [Holk <i>et al.</i> , 2024]	GPT-4	×	rew, o	single	single	online	π	rew, se, al	-
		ELLM [Du <i>et al.</i> , 2023]	Codex, GPT-3	×	rew, plan	single	multi	online	π_l, v_l	sr, gen, se, exp	-
		RL-VLM-F [Wang <i>et al.</i> , 2024]	Gemini-Pro, GPT-4V	×	rew, eval	single	single	online	π, v	sr, rew, se	Link
		VLM-RM [Rocamonde <i>et al.</i> , 2024]	CLIP	×	rew	single	single	online	π, v	sr, al	Link
		MineCLIP [Fan <i>et al.</i> , 2022]	CLIP	✓*	rew, eval	single	multi	online	π_l, v_l	sr, gen, se, al	Link

Taxonomy

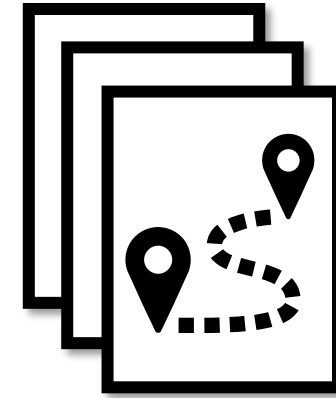


1

LLM/VLM as Agent



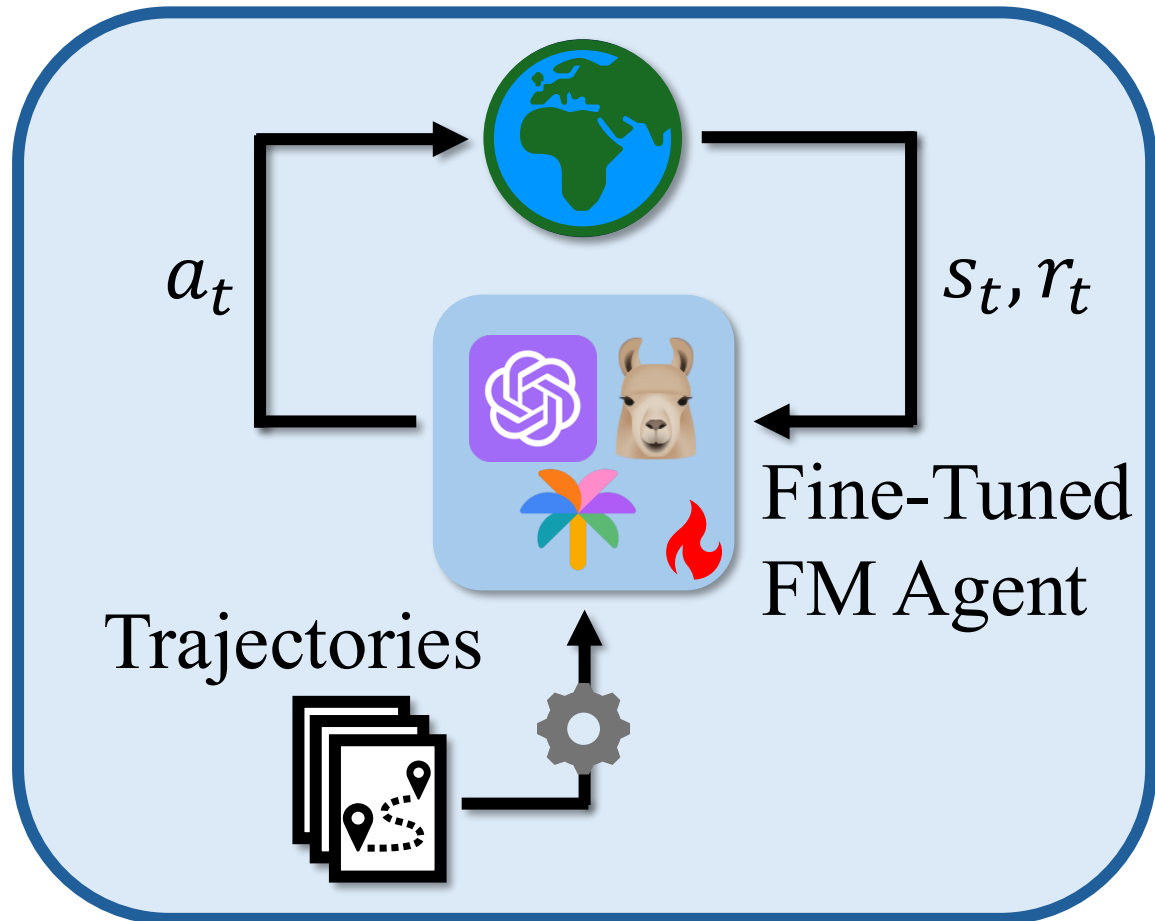
(Sub-)Trajectories



$s_0, a_0, r_1, s_1, a_1, r_2, s_2, a_2, r_3, \dots$

1

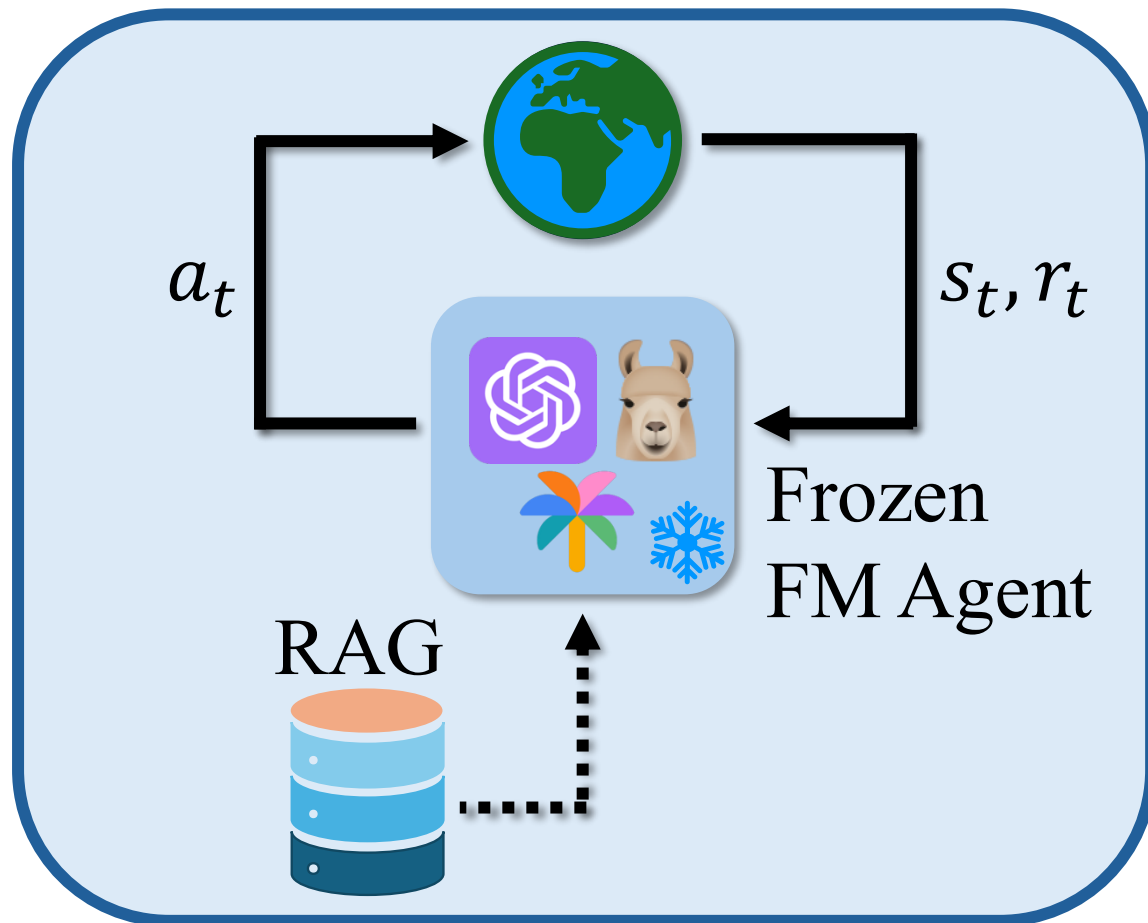
LLM/VLM as Agent: Parametric



- Parameter efficient fine-tuning techniques
 - LoRA
 - Adapters
 - Prefix tuning

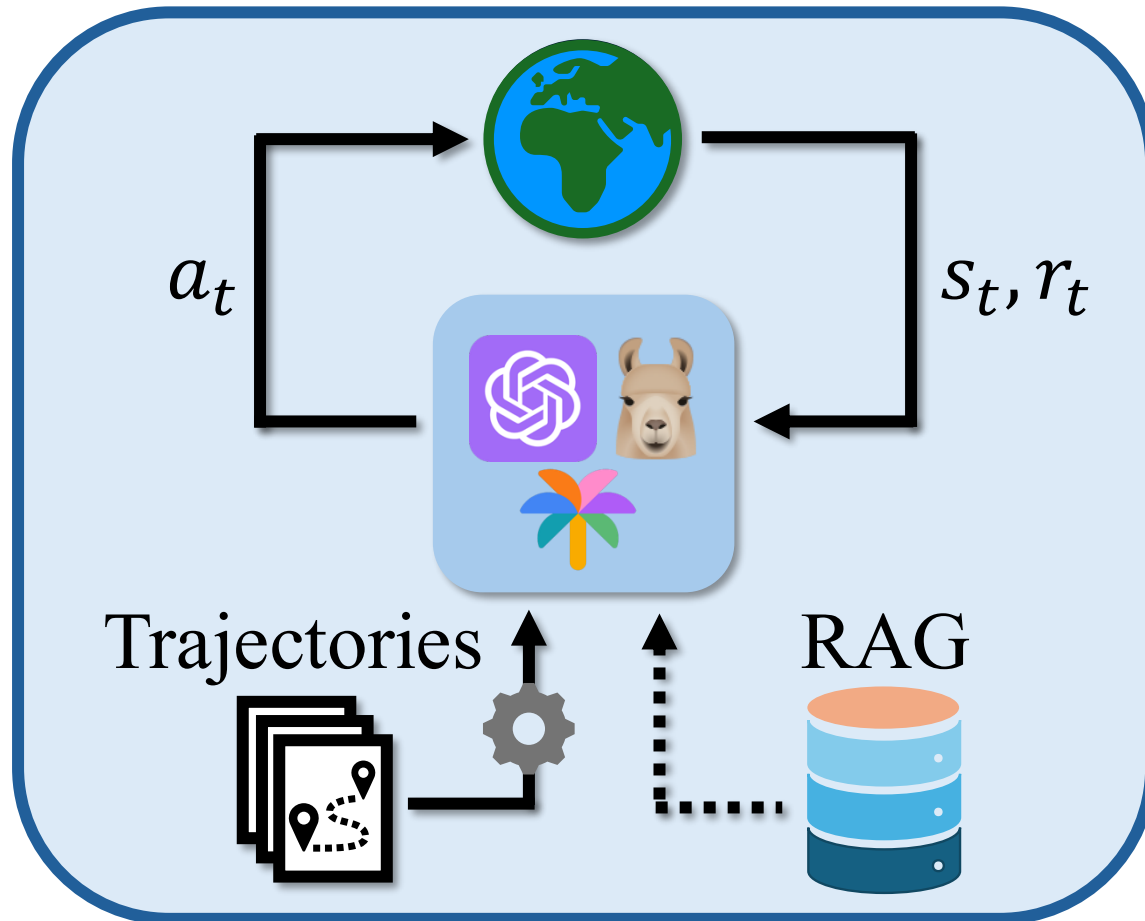
1

LLM/VLM as Agent: Non-Parametric



1

LLM/VLM as Agent



- Benefits

- Parametric: Retain broad capabilities but improve task-specific performance.
- Non-Parametric: No need for task-specific fine-tuning; improve performance through in-context learning.



AAAI2024

ExpeL: LLM Agents Are Experiential Learners

**Andrew Zhao¹, Daniel Huang², Quentin Xu², Matthieu Lin², Yong-Jin Liu²,
Gao Huang¹**

¹Department of Automation, BNRist, Tsinghua University, Beijing, China

²Department of Computer Science, BNRist, Tsinghua University, Beijing, China

Motivations

- LLMs excel at broad reasoning but struggle to retain task-specific experience
- Fine-tuning adds skills but is heavyweight and closed-source
 - Requires large task-specific data/compute
 - May degrade generalisation
- **Research Question:** Can an LLM agent autonomously learn from its own experience, boosting performance across tasks with no fine-tuning?

ExpeL (Inter-Task Learning)

Three phases:

- A. Experience Gathering
- B. Insights Extraction
- C. Inference

A. Experience Gathering

Reflexion [Shinn *et al.*, 2023]:
Intra-Task Learning

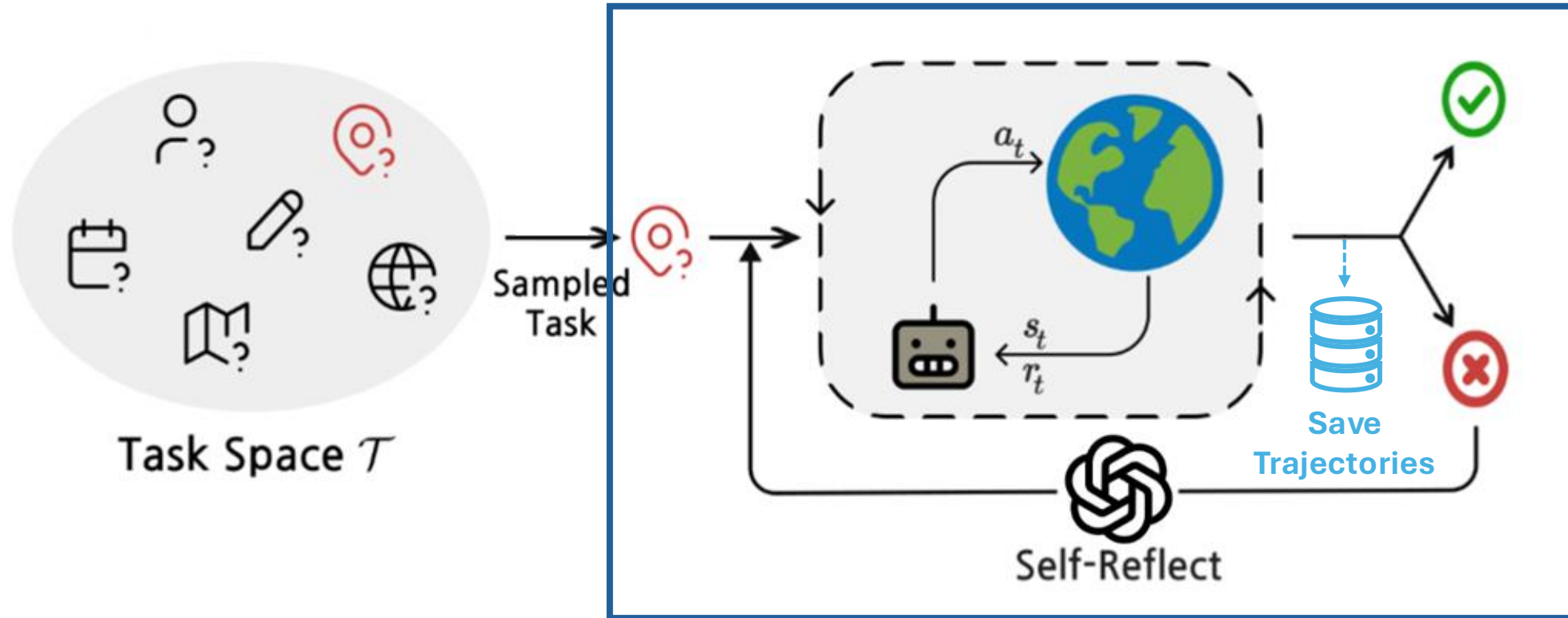


Figure adapted from Zhao *et al.* [2024].

B. Insights Extraction

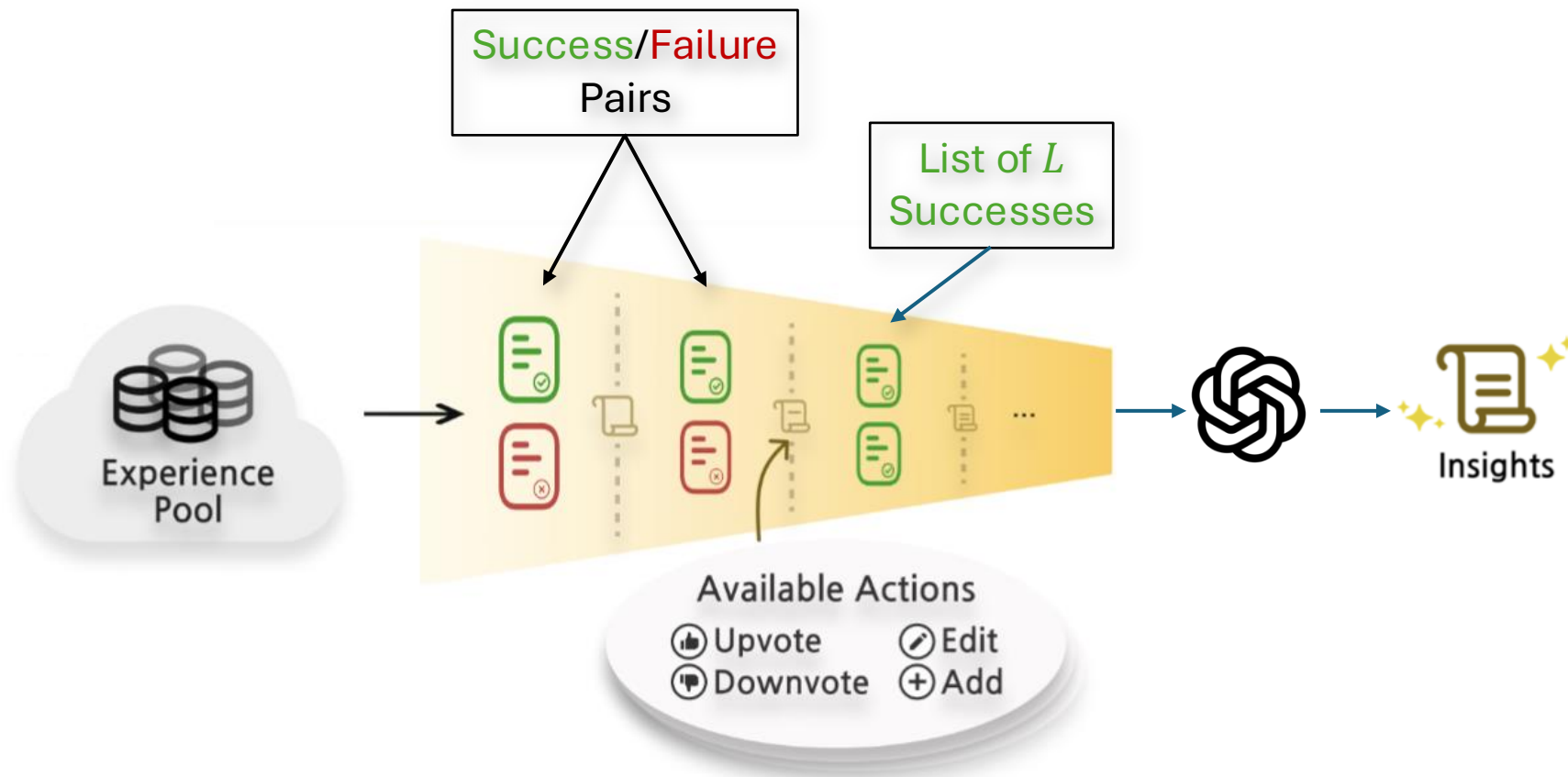


Figure adapted from Zhao *et al.* [2024].

C. Inference

- Extracted Insights used in-context as guidance.
- Retrieve in-context exemplars by extracting similar success experiences from the experience pool.

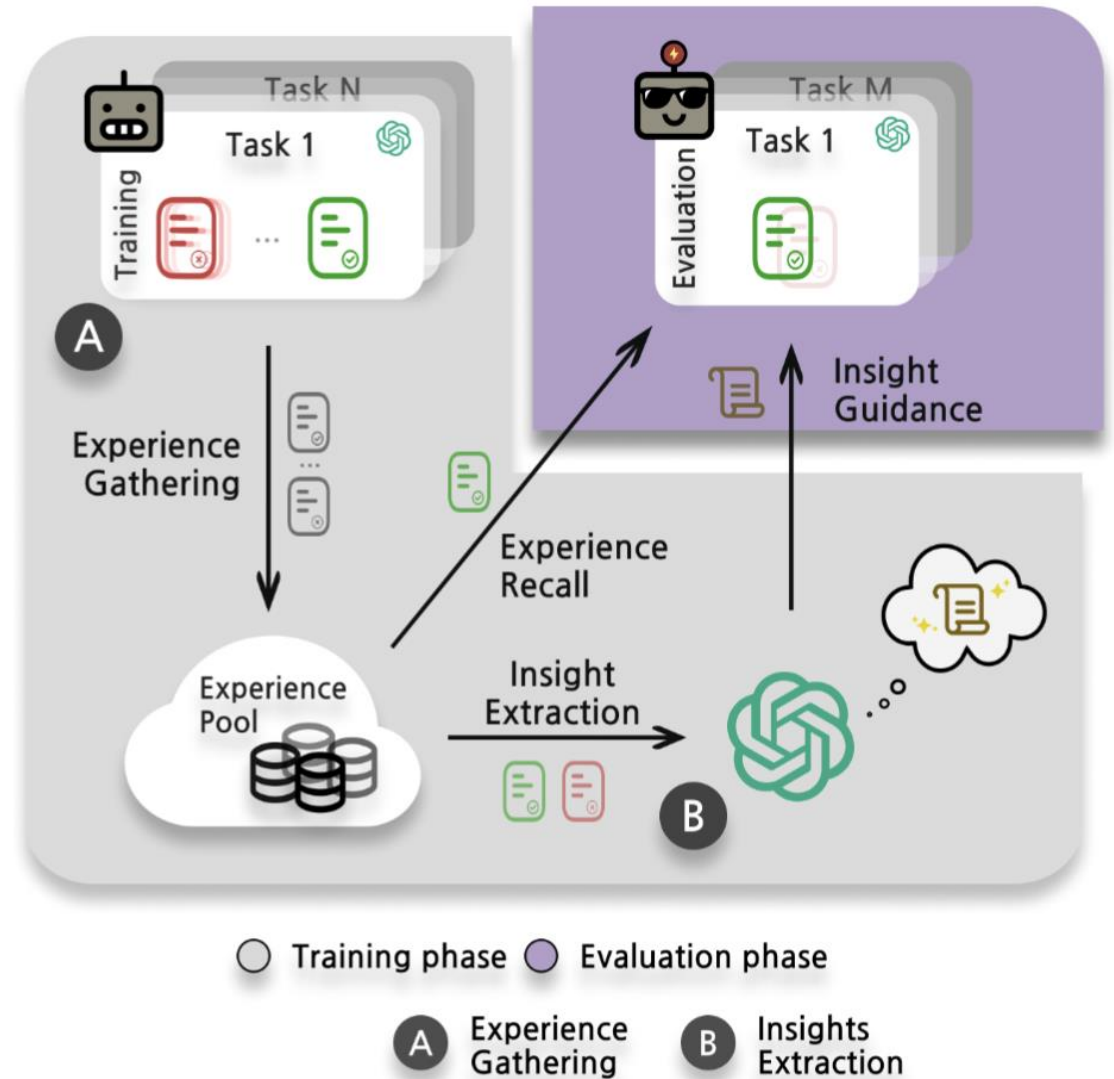


Figure adapted from Zhao *et al.* [2024].

Results

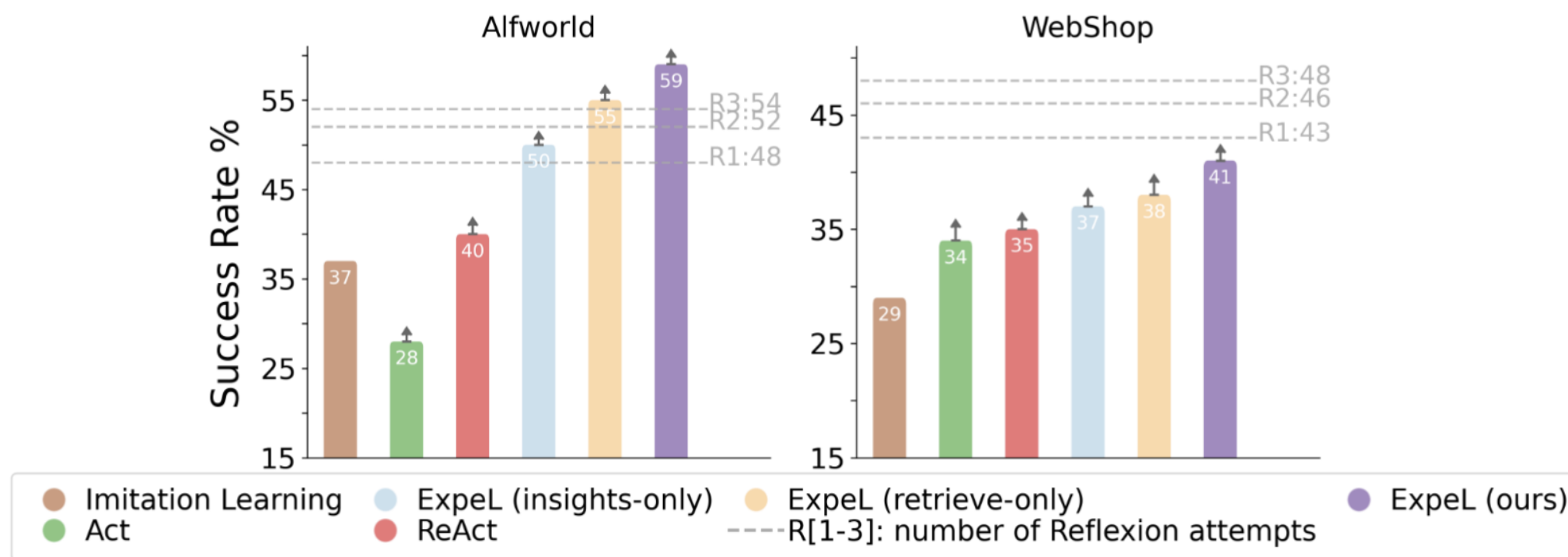


Figure adapted from Zhao *et al.* [2024].

2

LLM/VLM as Planner

Hey LLM! Get me a can of soda, *please*? 😊

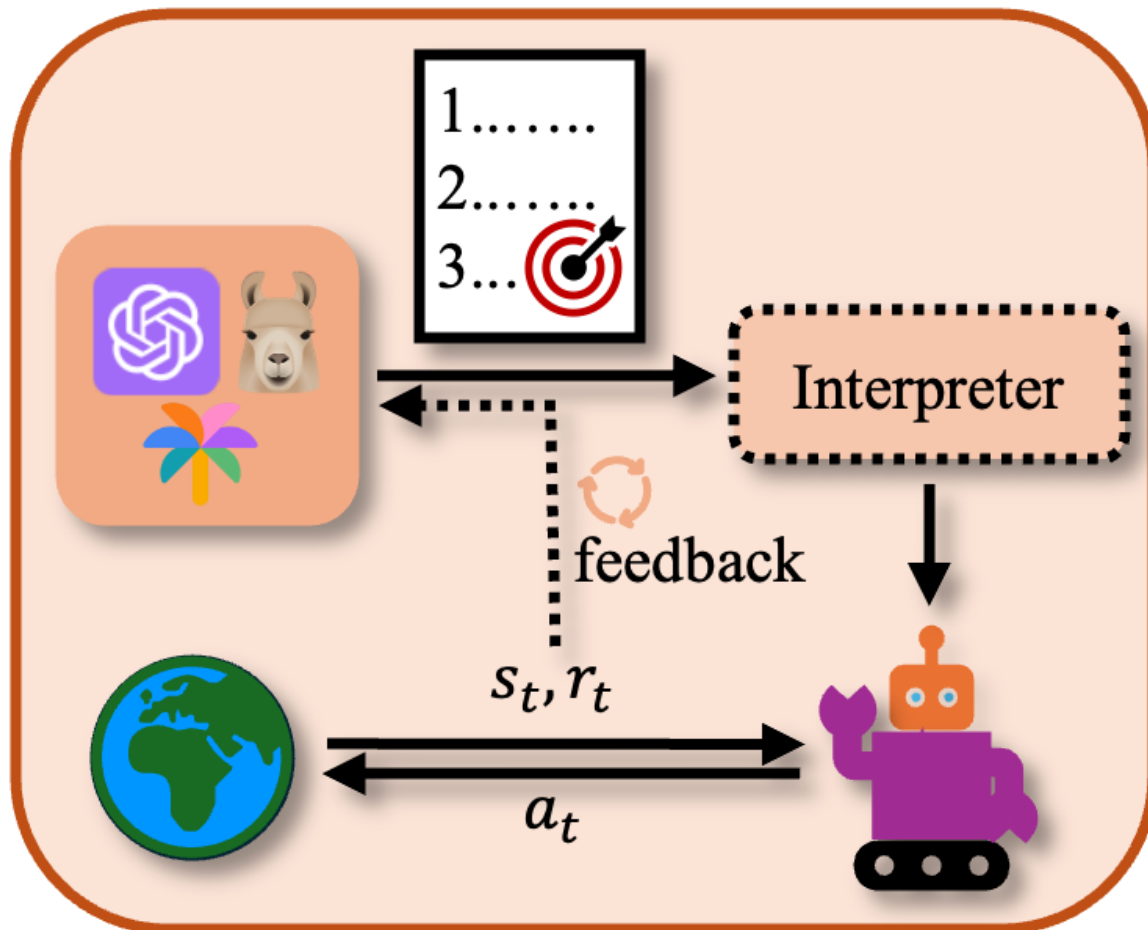


Sure thing! A plan to “get you a can of soda” is:

1. Locate soda can.
2. Navigate to <LOCATION>.
3. Grasp soda can.
4. Navigate to <PERSON>.
5. Place soda can.

2

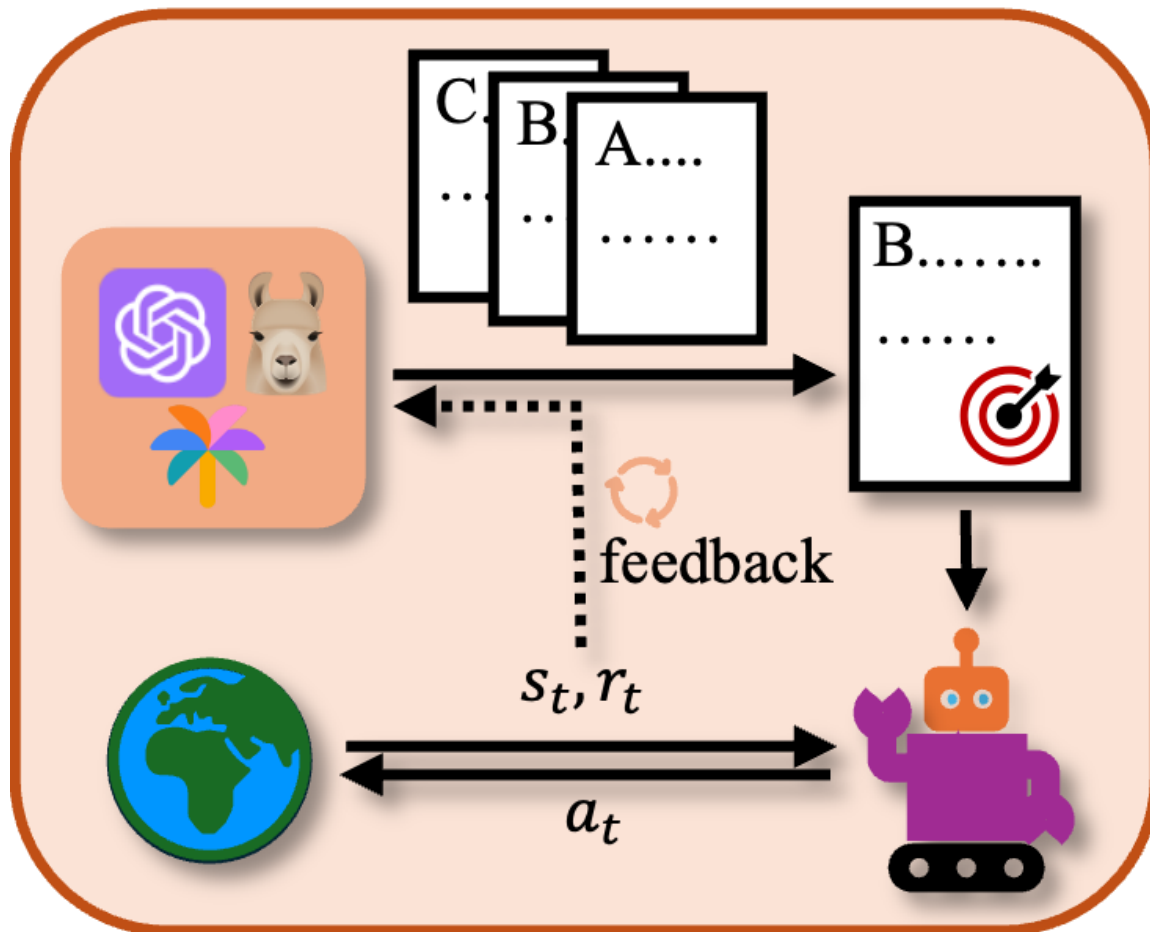
LLM/VLM as Planner: Comprehensive



- One query $\rightarrow$ full plan
- Interpreter translates each step of the plan into a goal that the agent can learn or execute
- Execution failure triggers full re-plan

2

LLM/VLM as Planner: Incremental



- One query $\rightarrow$ several next-step options
- Agent executes via language-conditioned policy
- Execution feedback triggers single-step re-planning on failure

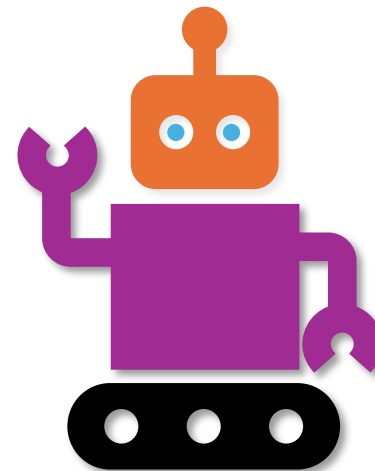
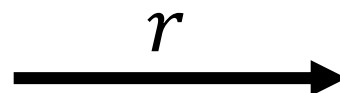
2

LLM/VLM as Planner

- Benefit
 - Break down long-horizon tasks into smaller sub-goals that are easier for the agent to execute or learn

3

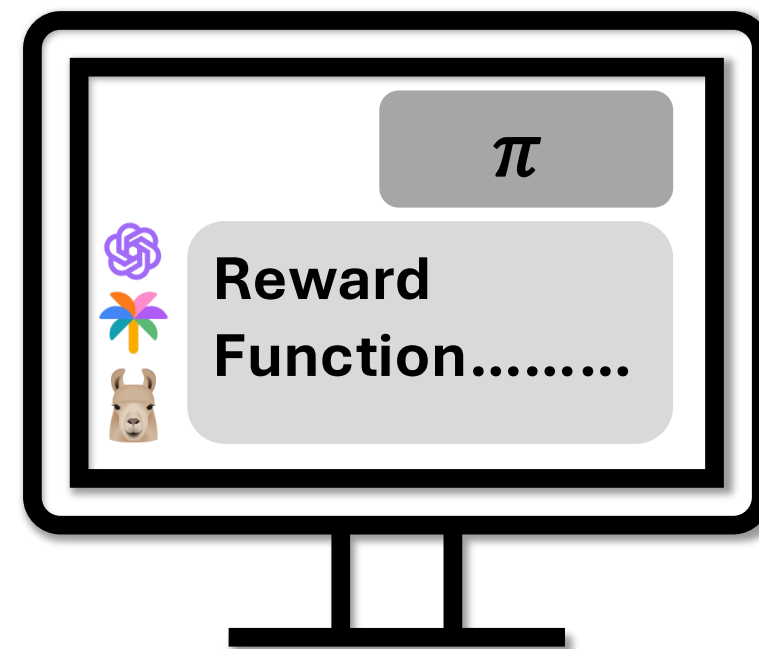
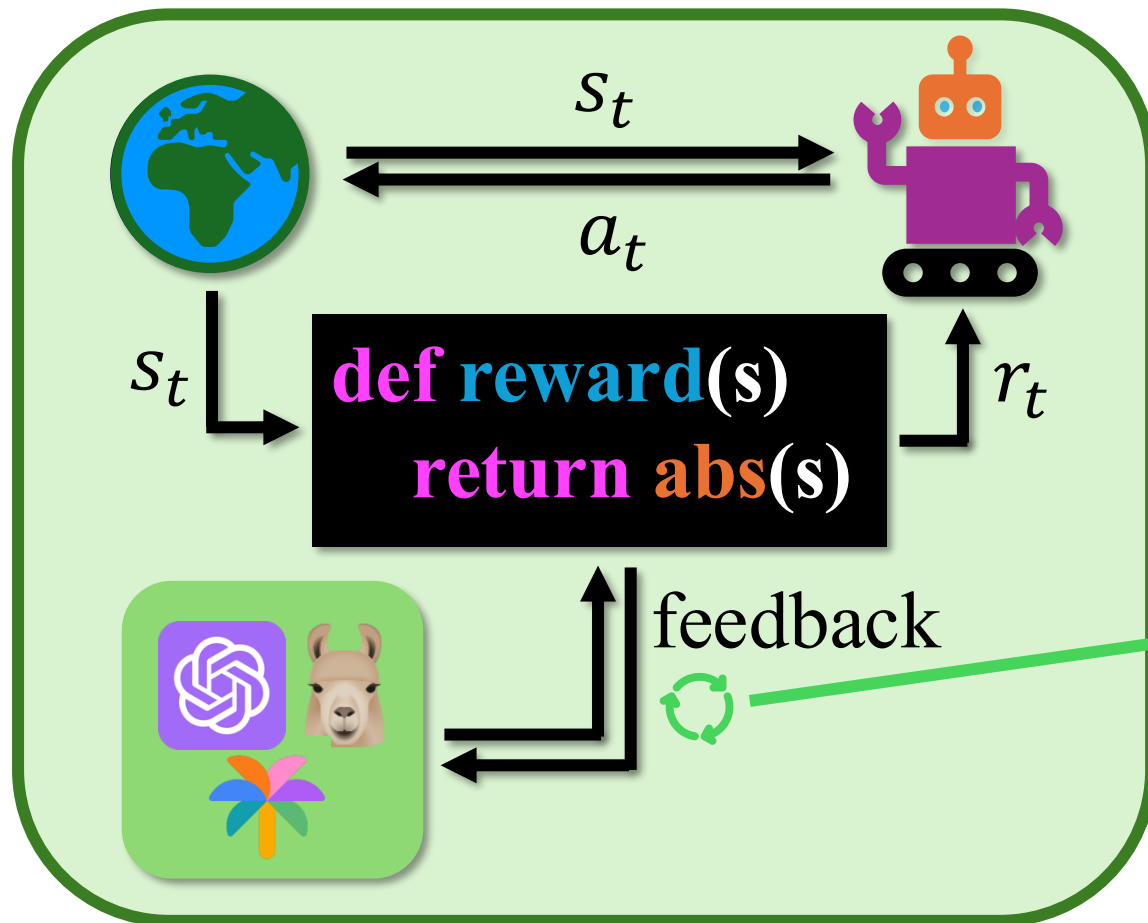
LLM/VLM as Reward



indirect / direct source of reward

3

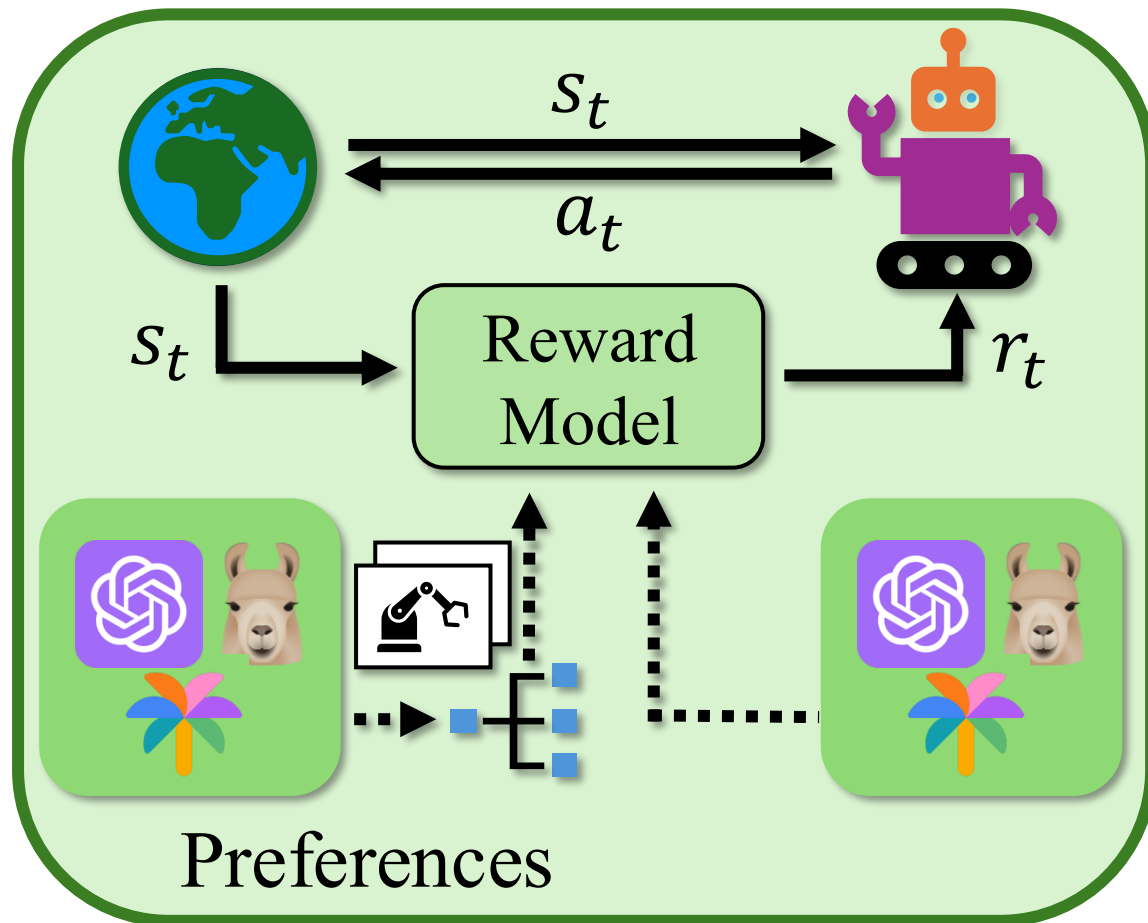
LLM/VLM as Reward: Function



In-Context Learning

3

LLM/VLM as Reward: Model



- Reward Model Learning (Left):
 - The LLM/VLM provides preferences (rankings) on trajectory sub-sequences for reward model learning
 - A learned reward model generates scalar rewards
- Reward Model (Right):
 - The LLM/VLM generates reward directly



LLM/VLM as Reward

- Benefits
 - Minimizes human efforts
 - Reward engineering
 - Providing preferences for reward model learning
 - Can learn reward functions that outperform those designed by humans



ICLR 2024

Eureka: Human-Level Reward Design via Coding Large Language Models

Yecheng Jason Ma^{1,2}, William Liang², Guanzhi Wang^{1,3}, De-An Huang¹,
Osbert Bastani², Dinesh Jayaraman², Yuke Zhu^{1,4}, Linxi “Jim” Fan¹, Anima
Anandkumar^{1,3}

¹NVIDIA

²UPenn

³Caltech

⁴UT Austin

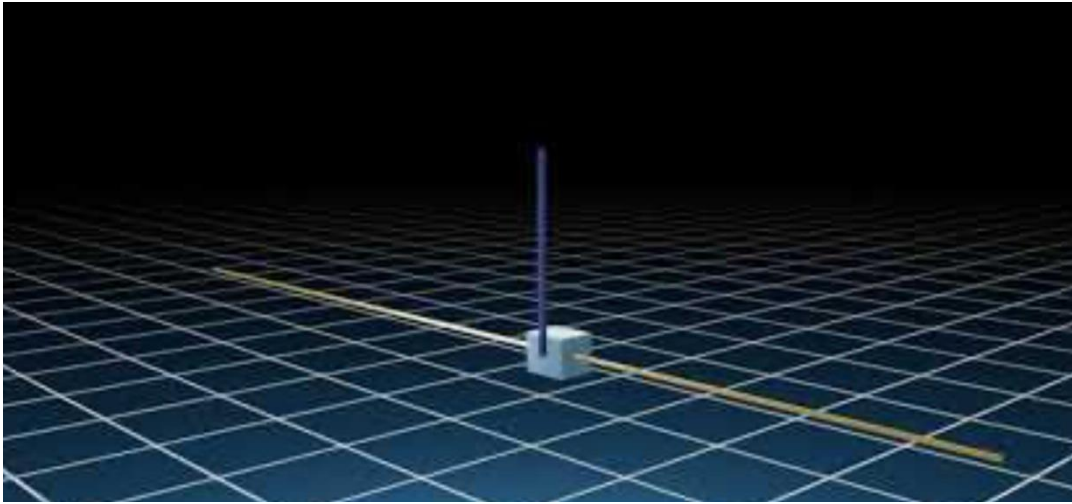
Motivations

- LLMs are good at high-level planning but struggle with low-level skills
- RL can master fine-grained skills, but requires carefully crafted reward functions
 - Trial-and-error search process
 - May be sub-optimal
- **Research Question:** Can a coding LLM autonomously find good reward functions?

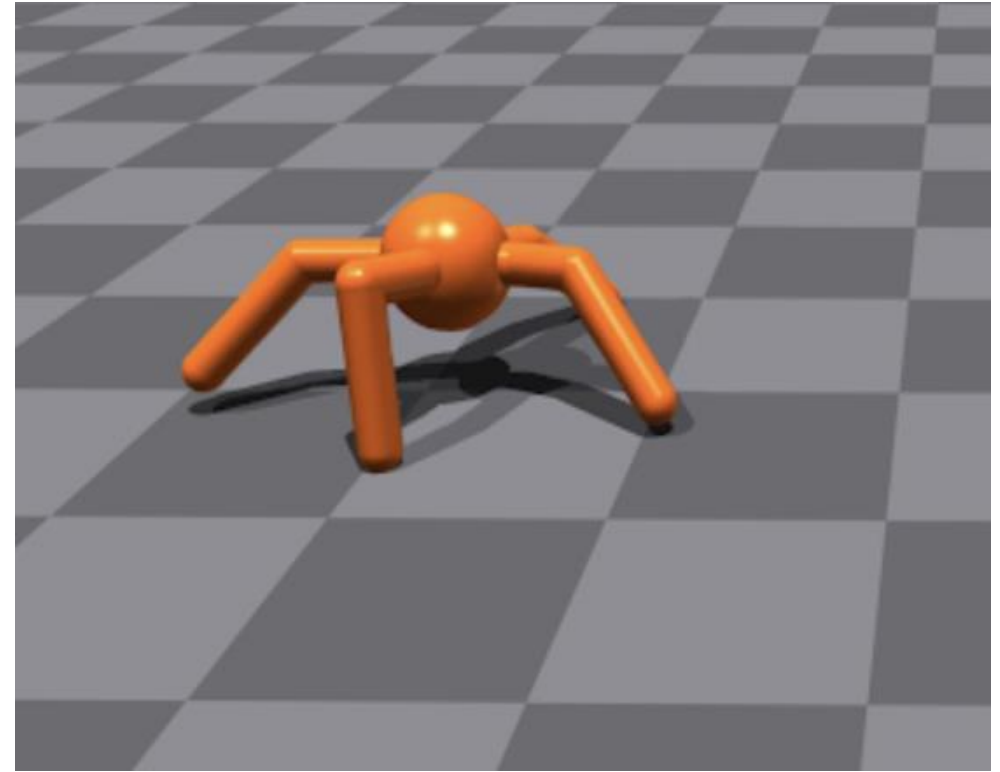
Reward Design Problem

- World model $M = (\mathcal{S}, \mathcal{A}, T)$ with states, actions, and dynamics
- $\mathcal{R}$ is the space of possible reward functions
- $A^M(\cdot)$ is a learning algorithm that maps any reward R to a policy π
- $F(\pi)$ is a scalar task metric (fitness) evaluating a policy π
 - Human-designed
 - Task-specific
 - Black-box to the LLM+RL system (can only be queried, not inspected)
- **Objective:** Find a reward function $R \in \mathcal{R}$ so that the learned policy π maximizes $F(\pi)$.

Fitness Function



Cartpole: duration (how long the pole stands up)



Ant runner: $\text{cur_dist} - \text{prev_dist}$ (forward progress per step)

Eureka: Evolutionary Search

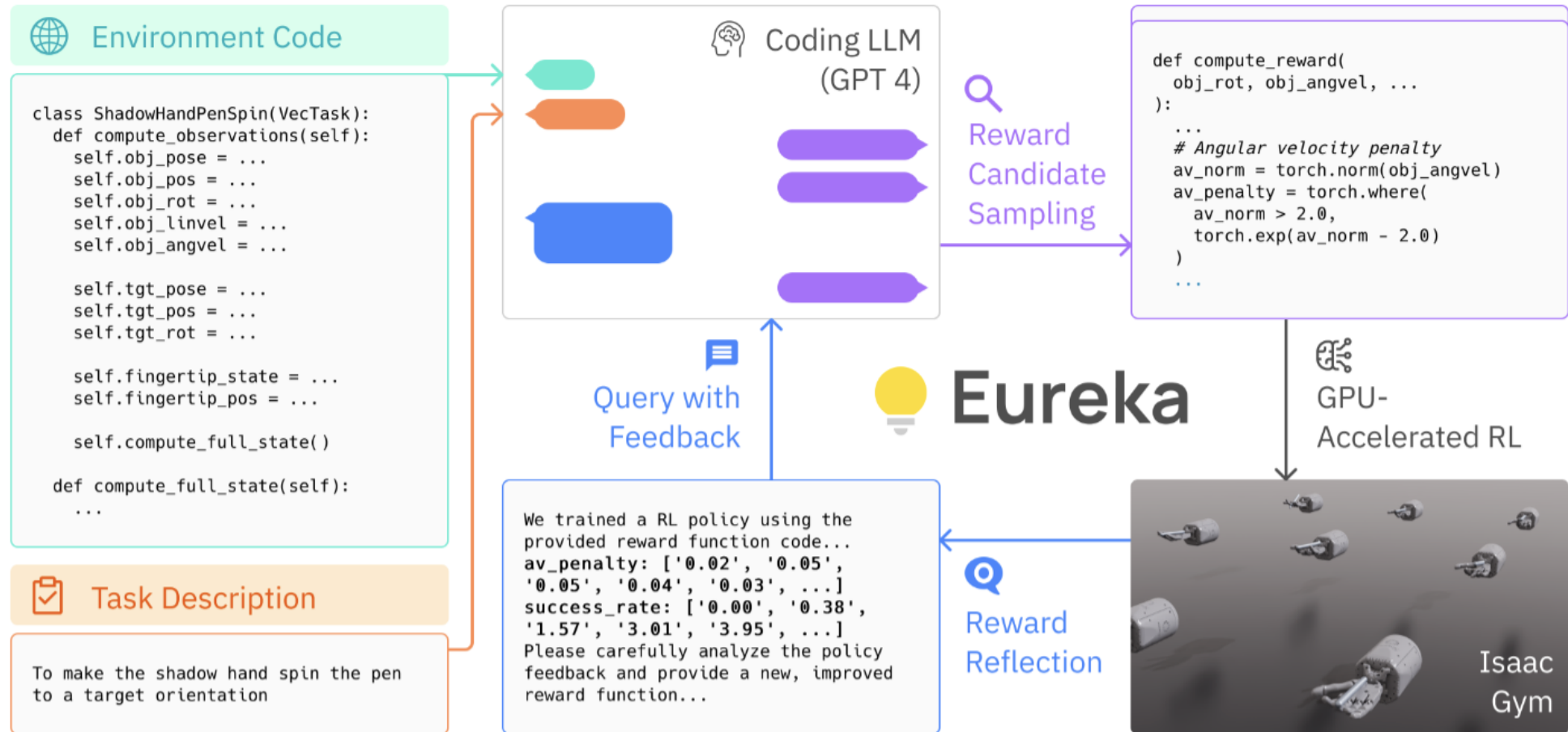


Figure from Ma *et al.* [2024].

Improvements Over Baselines

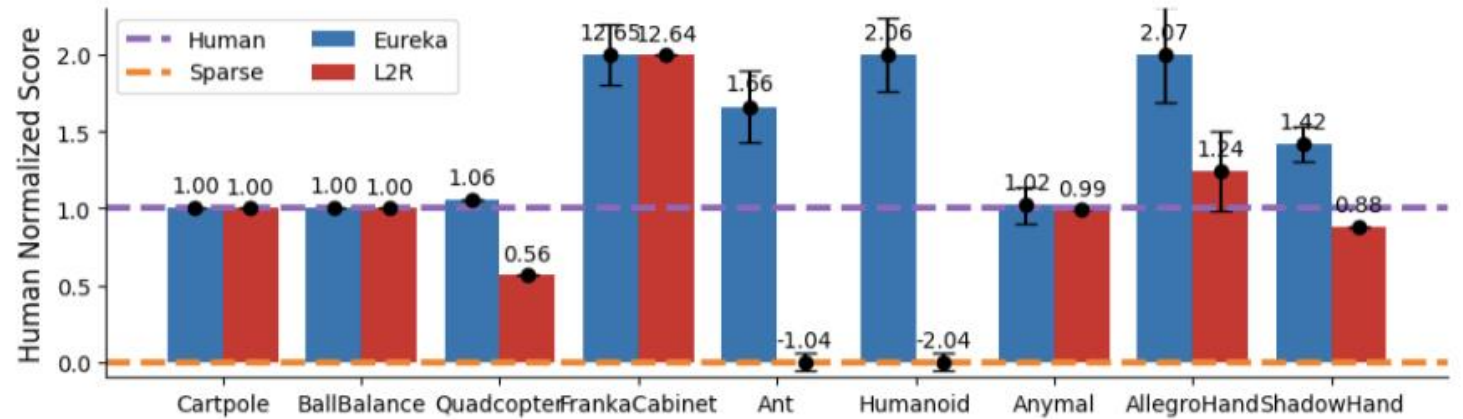
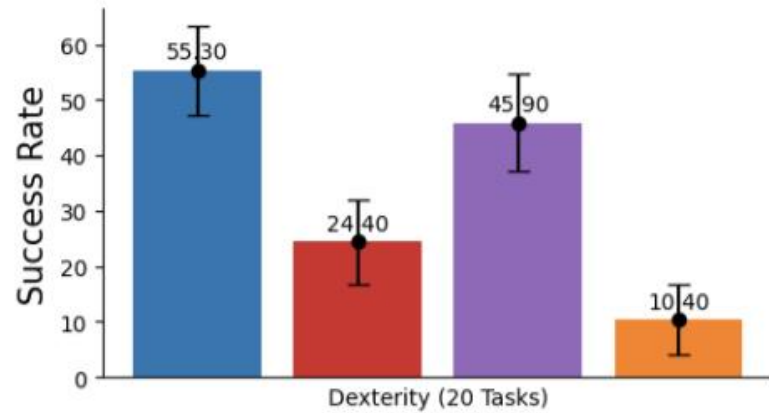


Figure from Ma *et al.* [2024].

Improvements Over Iterations

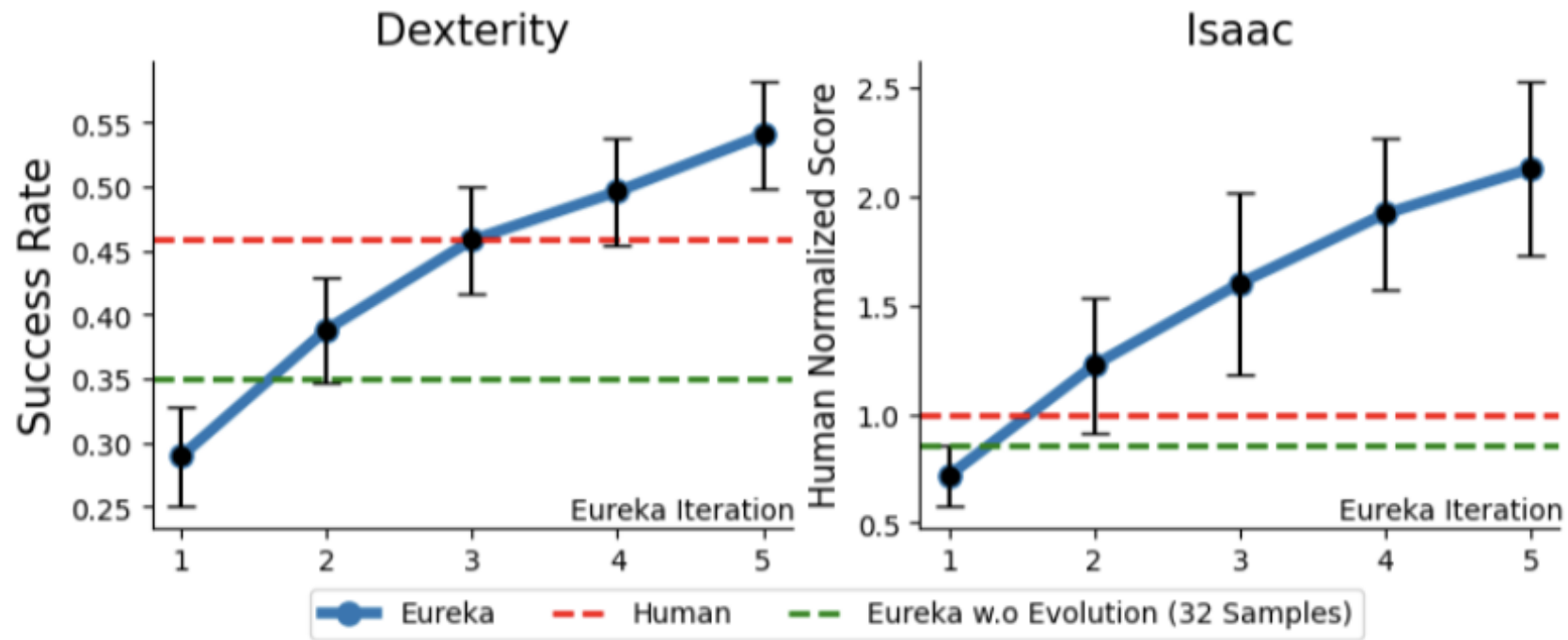


Figure from Ma *et al.* [2024].

Future Directions

Future Directions: Representation (LLMs)

Problem

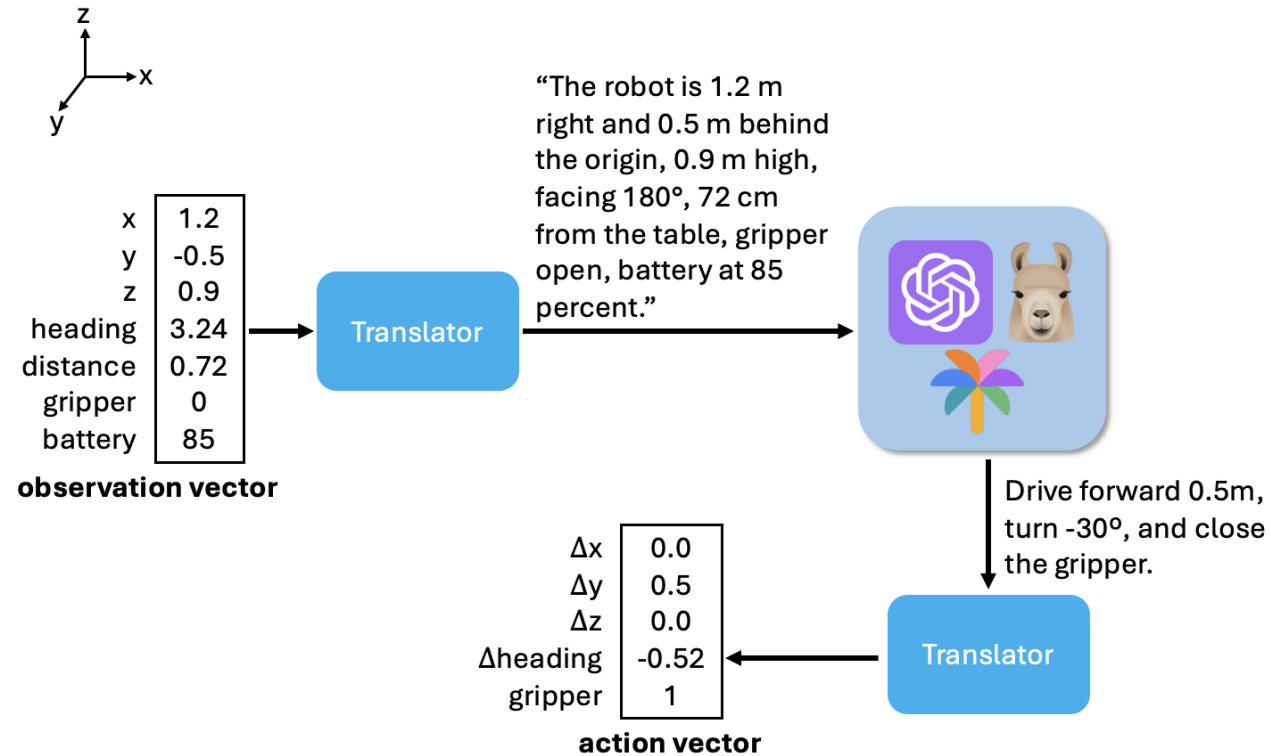
- Converting rich numeric vector representations to tokens and back

Where we stand

- Text-based environments
- External translator modules
- Architectural modifications

Gaps

- Latency and information loss
- Integration of VLMs



Future Directions: Grounding

Problem

- Steps in a plan generated by an LLM/VLM may be infeasible in practice.

Where we stand

- External modules validate feasibility
- Task-specific output formats (e.g., binary foot contact patterns)

Gaps

- Limited generalization
- Bias can misdirect learning



Pick up the apple.

Future Directions: Action Advice

Problem

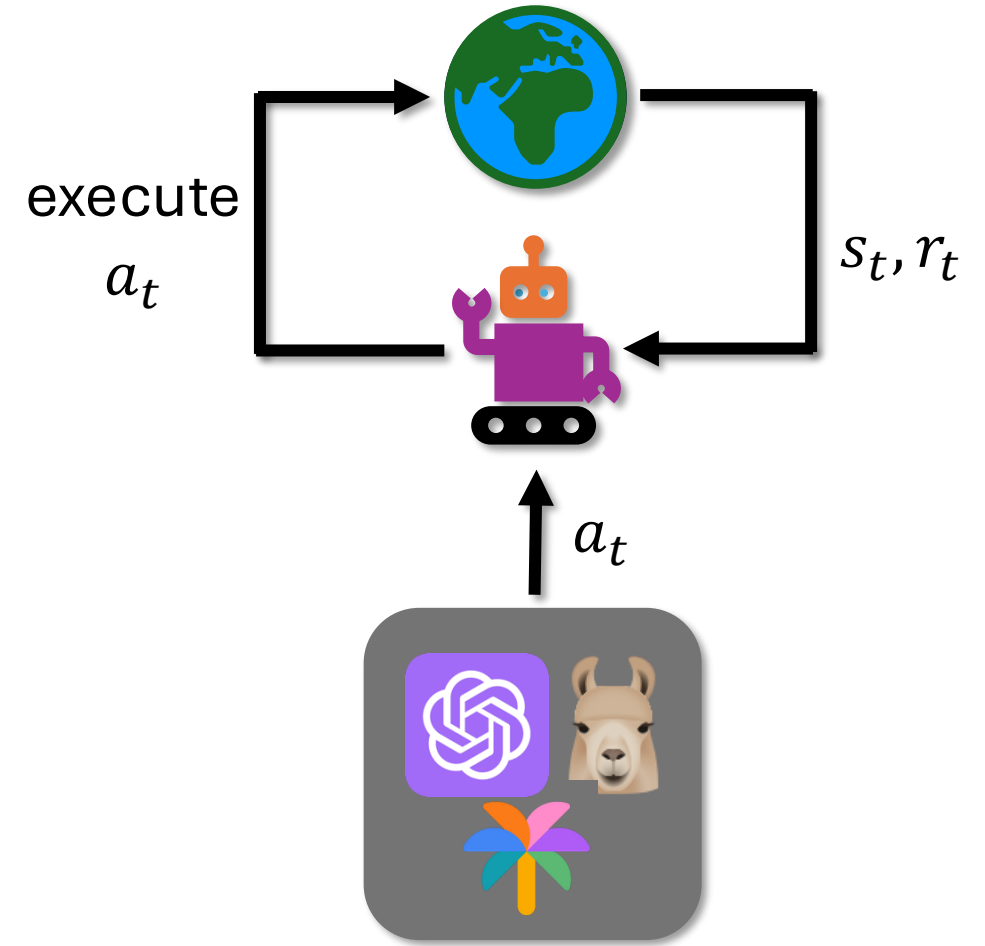
- RL learns slowly

Where we stand

- Expert teachers (human/artificial) accelerate student agent learning

Gaps

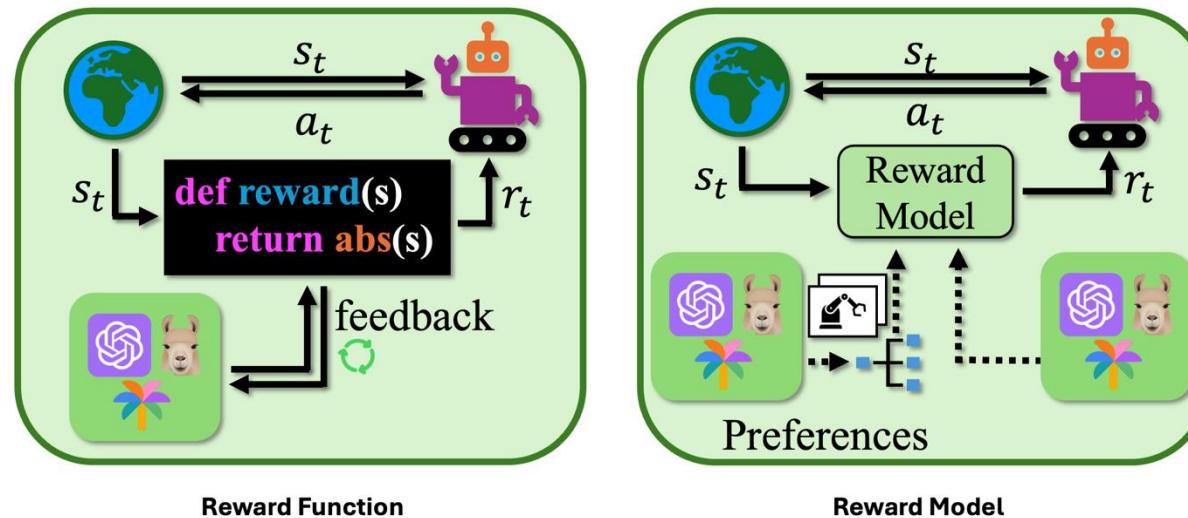
- LLMs/VLMs serve as a teacher





Key Takeaways

3 LLM/VLM as Reward



Questions?

References

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